Onigroup SOW - Class and Homework Assignment

Class Set Up & Assignment Flow

Version 1.1 | March 2026

Version 1.0 | February 2026

Teacher Flow: [Teacher View for AI Grading – Figma Make]

Student Flow: [AI Grading (Student View) – Figma Make]

Designs: figma.com/design/BZbdZoiyL0m0HCfF6HdOBd/Homework-Assignment?node-id=29-2035

Examples of Submissions
Mini Test (MAIN USE CASE): 小テスト - JP - Google Drive
Assigments (SECONDARY USE CASE): Examples of HW to Mark - Google Sheets

1. Problem Direction

Japan's juku market is characterised by high teacher workload, workbook-based homework collection, and fragmented feedback loops. Critically, it has been reported that some staff take more than 20 days to return homework to students, a delay that directly undermines the learning loop homework is meant to create.

Teachers currently lack a scalable, easy mechanism to distribute assignments, track submissions, and deliver marked feedback at pace. This production-ready prototype addresses three validated pain points from juku teachers:

- No easy entry point for teachers to onboard students into a class and push assignments digitally.
- Homework marking at scale is manual, creating severe bottlenecks — feedback can take 20+ days to reach students.
- Students cannot act on feedback effectively — by the time manual marking is returned the feedback is rarely specific enough to drive improvement.

This feature scope creates a webview-based teacher onboarding and assignment flow that integrates with Manabie's existing LMS (tenant → location → course → class) and AI marking infrastructure — validating whether a lightweight, QR-code-based class model can drive teacher and student adoption at the top of the funnel.

2. Goals & Non-Goals

2.1 Goals

- Validate teacher activation: can a new teacher sign up, create a class, and issue an assignment in under 5 minutes.
- Validate student self-enrolment via QR code scan — zero admin overhead for the teacher.
- Confirm AI marking integration: does the existing marking engine surface correctly for photo-submitted homework?
- Generate qualitative adoption signal: will teachers adopt AI marking if onboarding friction is below a threshold?

2.2 Non-Goals (V1)

- Teacher's landing page.
- No Learnosity integration — student feedback will be webview only.
- No multi-school admin portal or school-level analytics dashboards.
- No native iOS/Android screens — student submission is mobile webview only.
- No bulk assignment across multiple classes simultaneously.
- No CSV or roster import for students - student management handled by
- No multi-teacher class co-ownership.

3. Success Metrics

The prototype is considered validated when all Primary metrics are met. Secondary metrics inform the decision to productise.

Metric	Target	Measurement Method
PRIMARY — ADOPTION & ACTIVATION		

Teacher signup → first assignment created	≤ 5 min	Event timestamps in analytics pipeline
Teacher 7-day retention (≥1 class active)	≥ 65%	Active class with ≥1 student in week 2
Student join rate via class link	≥ 70%	Join events / link share events
Assignment submission rate	≥ 50%	Submissions / assignment recipients
SECONDARY — QUALITY & SATISFACTION		
Teacher NPS (post 2-week pilot)	≥ 30	
Assignment creation to mark turnaround	≤ 24 hrs	

4. **User Flows**

The end-to-end flow spans these actors: Teacher and Student.

Teacher Flow

- Sign up → select school (school selection by Sales picklist value or staff / existing: add to staff)
- Create class → class ID generated → QR code displayed + share QR code
- Create assignment → set instructions → configure student self-marking toggle
- Review submissions → view AI marking → edit if needed → approve and release

Student Flow

- Student scans QR code shared by teacher → 'Scan Class QR Code' screen → 'Class Joined!' confirmation
  - Alternative: tap 'Add Class' → enter class code manually
- Student sees assignment in homework feed
- Student taps assignment → Assignment Details → taps 'Submit Your Work'
- 'Take or Upload Photo' options → student photographs work (4 pages max)

- Preview screen → taps 'Confirm & Analyse' CTA
- 'Analysing Your Work...' loading state while OCR + AI marking engine processes submission
- 'Analysis Results' screen: overall score + question-level breakdown score
  - Student can remark
- Student taps 'Add AI to Teacher' or 'Submit to Teacher' → submission sent
- Homework feed item updates to 'Submitted'; after teacher approval: 'Marked'

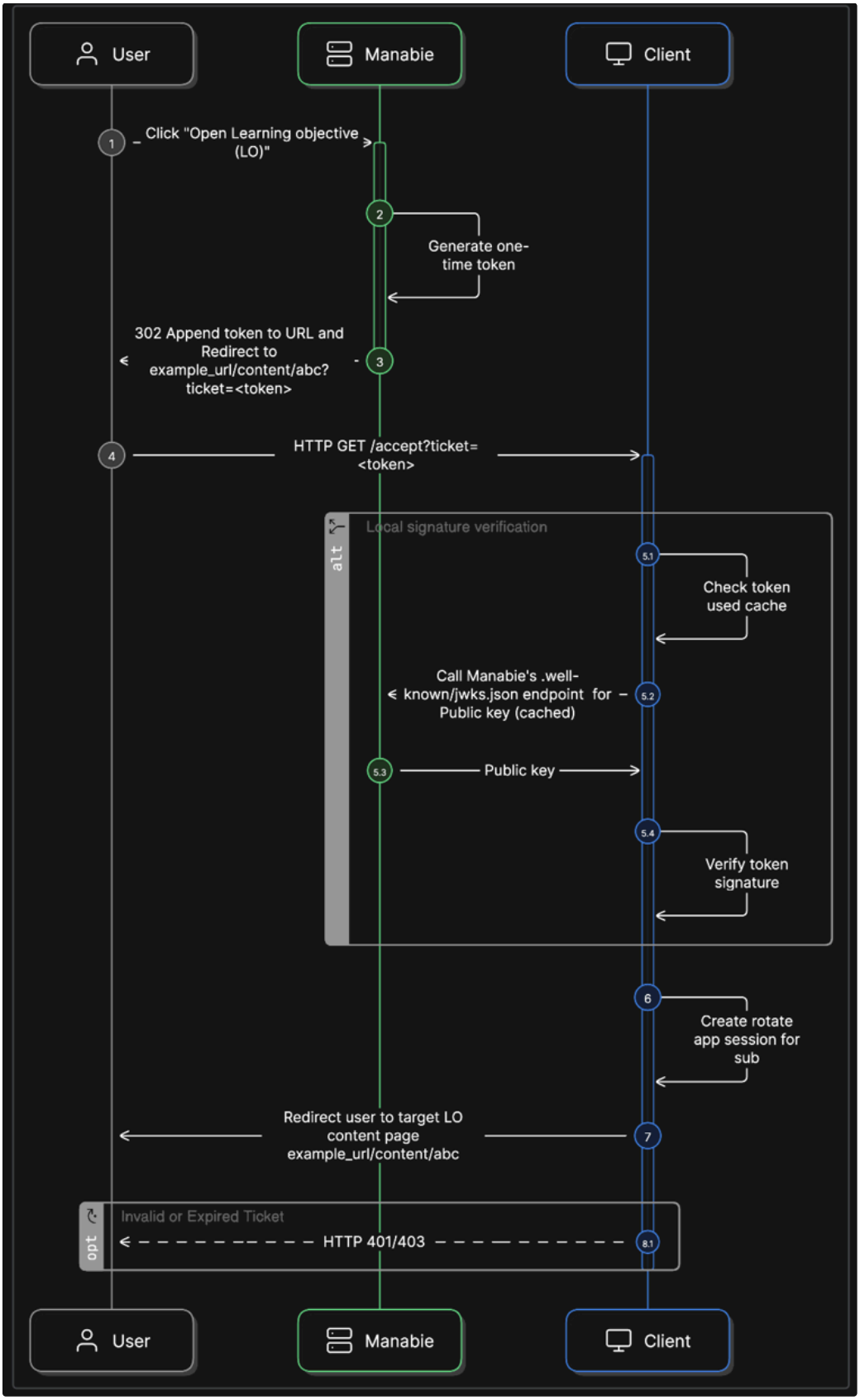
5. Tech Stacks

UI: React to embed as a WebView
BE: Python (raw Python or Langchain to connect to Gemini)
Models: Gemini

5.1 Technical Built

- Registration flow:
  - Handled by Manabie
- LOGIN — embed into BO so that login flows reside with Manabie
- Register landing page for teacher — Manabie builds this, calls teacher API and sends email login details
- Student join flow
  - Students join via QR code scan
  - QR code encodes the class join URL; scanning opens the webview directly into class enrolment
  - Google Identity Platform integration required — link must be cap-sensitive and not guessable
- After login from webview (student):
  - Class menu displayed for marking tenants
  - When student taps class → opens Oni webview
  - Class ↔ student relationship handled by Oni
  - Assignment ↔ student relationship handled by Oni
  - Student submits assignment in webview; sees assignment stats in webview
- Manabie handles authentication to open webview (passing teacher_id for BO, student_id for learner)
- Onigroup will improve OCR to capture questions and student answers and send to Manabie AI Feedback API

5.2 Manabie & Oni System Integration Spec (Asymmetric JWT)



6.0 SOW

Week	Activity	What is Required
PHASE 1 — Mobilization & Environment Setup		
1	Tech/Spec Understanding, Deployment Pipelines, UX Discussion. Setting up evaluation workflow. Auth Design: Design student QR code join flows. Define API contracts: Onigroup Dev API spec draft. Wireframe review.	—
PHASE 2 — Teacher Set Up — Class, HW Creation, Marking Flow		
2–5	AI Marking Engine Critique and Improvement. Class creation with QR code generation. Teacher opens submissions, edits feedback and scores, and returns.	Marking Agent API, Submission-level Feedback Agent API
PHASE 3 — Student Set Up (WebView with mobile optimisation)		
6–9	QR code scan flow for student class join. Assignment creation flow linked to student	OCR API

	ingestion, submission and display.	
PHASE 4 — Insights & Handover		
10–12	Start Insights Tool. Optimisation, evaluation and handover. Transfer Code.	Sign Off

7. User Stories & Functional Requirements

User Story

"As a teacher, I want to create a class under my school and receive a QR code I can share with students — so they can join instantly without needing to create an account."

Acceptance Criteria

- Teacher clicks 'Create Class' and enters: Class Name, Subject.
- Class Name is 100 characters max (text input). Subject is a free-text field.
- System creates a class record tied to the teacher's location and staff profile.
- A uniqu Class ID is generated and displayed on the class detail view.
- Class detail view displays a scannable QR code + a ‘Share QR Code’ button.
  - QR code encodes: https://[domain]/join/[class_id]
  - ‘Share QR Code’ opens the device share sheet so the teacher can send the QR image via any channel (messaging, email, etc.).
- Teachers can only view and manage classes they own in this webview.
- Class card on dashboard shows: class name, subject, student count, ‘Manage Class’, and ‘Share QR’ buttons.

7.2 Student Class Join via QR Code

User Story

"As a student, I want to scan my teacher's QR code so that I am immediately enrolled in the class — without needing to create an account or remember a password."

Acceptance Criteria

- Student scans QR code (via camera app or in-app scanner) → webview opens to /join/[class_id].
- 'Scan Class QR Code' screen shown during scanning; on successful scan: 'Class Joined!' confirmation displayed with class name.
- Alternative path — Manual Code Entry ('Add Class' flow):
  - Student taps 'Key Icon' → enters class code manually.
  - Step 1: Enter class code → Step 2: 'Class Joined!' confirmation.
- On successful join, student is immediately enrolled; assignment feed populates.
- If class code is invalid → error state: 'Class not found. Please check the code with your teacher.'

7.4 Assignment Creation

User Story

"As a teacher, I want to create an assignment with instructions and configure submission settings, so that it appears in my students' homework feed."

Acceptance Criteria

- Teacher clicks '+ New Assignment' from the class management view. A modal opens with the assignment title, uploaded files section, and question configuration panel.
- Required field: Assignment title / instructions (free text).
- File uploads: teacher can attach one files to the assignment.
  - Uploaded files are displayed in the modal under 'Uploaded Files' with file name and a 'View' link each.
  - Uploaded files are accessible to students from the assignment detail view (shown as 'View files (N)' link).
- Question configuration: teacher defines questions in a left-panel list (Question 1.1, 1.2, 1.3...). For each question:
  - Marks: numeric input field shown inline next to question number (e.g. 'Question 1.1 [1] marks').
  - Question text: free-text field describing the question.
  - Marking guide: free-text field for the correct answer used by the AI marking engine
  - Left panel shows question list with marks and truncated question text. Clicking a question loads it in the right panel for editing.
- On 'Assign', the assignment is pushed to all currently enrolled students

- Students who join the class after assignment creation can still see all current (non-expired) assignments.

User Story

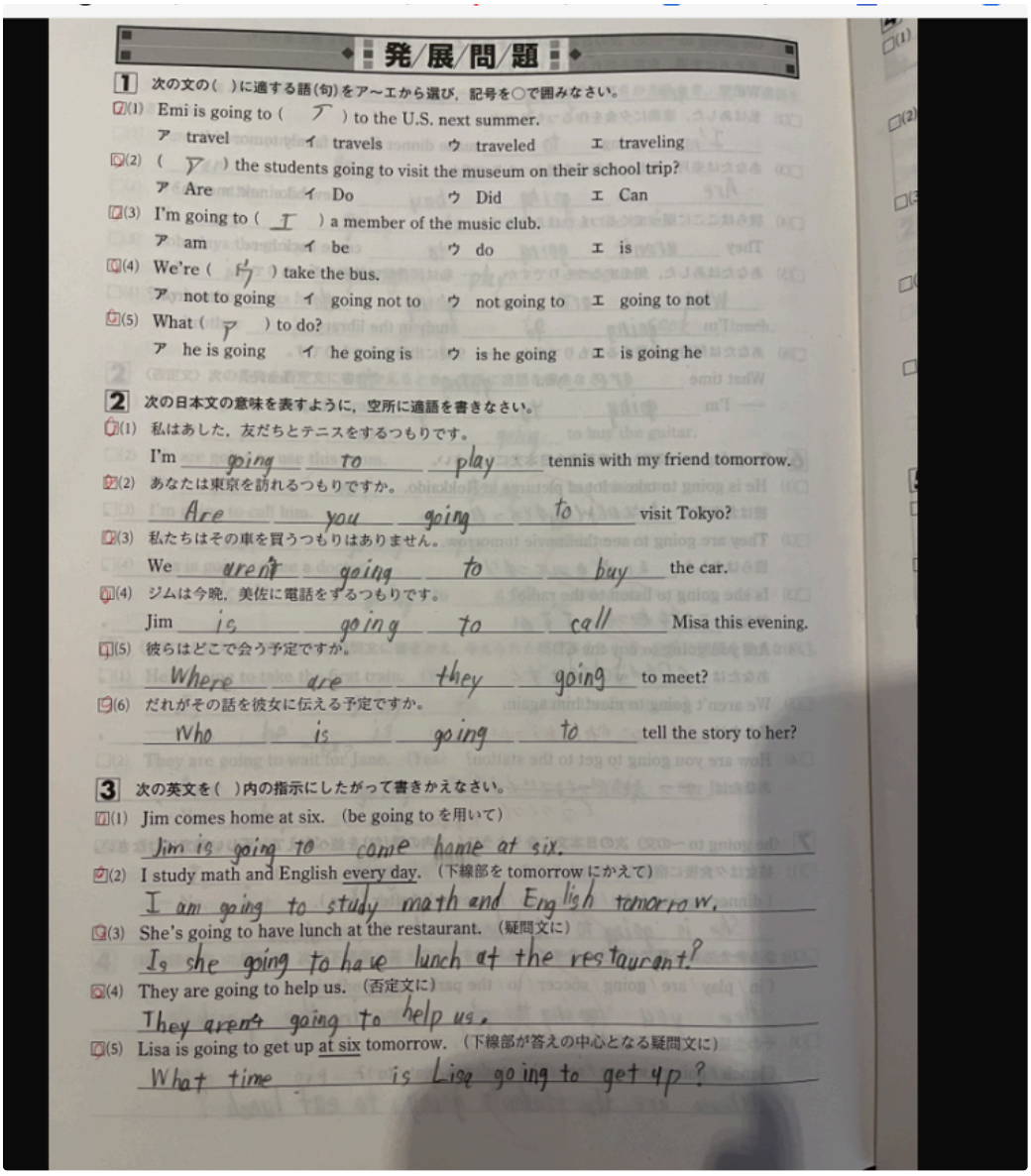
"As a student, I want to photograph my completed homework, review the AI analysis results, and then submit to my teacher — with clear confirmation at each step."

Acceptance Criteria

- Student opens Additional Homework item → sees Assignment Details screen with instructions and 'Submit Your Work' CTA.
- Tap ‘Submit Your Work’ → three options presented: ‘Take Photo’, ‘Upload File’

Acceptance Criteria

- Student opens Additional Homework item and sees assignment instructions.
- Tap "Submit" → device camera opens. Student photographs completed work.
- Student can upload up to 4 photos per assignment. All images are submitted together as a single submission and processed by the AI marking engine as one set.
- Preview screen shows thumbnails of uploaded images or file previews. Student can retake or re-upload individual pages before confirming.
- Student taps 'Confirm & Analyse' CTA.
- 'Analysing Your Work...' loading state displayed while OCR and AI marking engine processes submission.
- 'Analysis Results' screen displayed:
  - Overall score (e.g. 4/5).
  - Question-level breakdown: each question listed with a score. student can remark individual questions (override AI mark) before submitting to teacher.
- Student clicks 'Submit to Teacher' CTA → submission sent to teacher review queue.
- Homework feed item updates to 'Submitted' with success visual.



◆ 発/展/問/題 ◆

1 次の文の( )に適する語(句)をア～エから選び，記号を○で囲みなさい。

(1) Emi is going to (ア) to the U.S. next summer.
ア travel　イ travels　ウ traveled　エ traveling

(2) (ア) the students going to visit the museum on their school trip?
ア Are　イ Do　ウ Did　エ Can

(3) I'm going to (エ) a member of the music club.
ア am　イ be　ウ do　エ is

(4) We're (ウ) take the bus.
ア not to going　イ going not to　ウ not going to　エ going to not

(5) What (ア) to do?
ア he is going　イ he going is　ウ is he going　エ is going he

2 次の日本文の意味を表すように，空所に適語を書きなさい。

(1) 私はあした，友だちとテニスをするつもりです。
I'm going to play tennis with my friend tomorrow.

(2) あなたは東京を訪れるつもりですか。
Are you going to visit Tokyo?

(3) 私たちはその車を買うつもりはありません。
We aren't going to buy the car.

(4) ジムは今晩，美佐に電話をするつもりです。
Jim is going to call Misa this evening.

(5) 彼らはどこで会う予定ですか。
Where are they going to meet?

(6) だれがその話を彼女に伝える予定ですか。
Who is going to tell the story to her?

3 次の英文を( )内の指示にしたがって書きかえなさい。

(1) Jim comes home at six. (be going to を用いて)
Jim is going to come home at six.

(2) I study math and English every day. (下線部を tomorrow にかえて)
I am going to study math and English tomorrow.

(3) She's going to have lunch at the restaurant. (疑問文に)
Is she going to have lunch at the restaurant?

(4) They are going to help us. (否定文に)
They aren't going to help us.

(5) Lisa is going to get up at six tomorrow. (下線部が答えの中心となる疑問文に)
What time is Lisa going to get up?

Other Examples: [Examples of HW to Mark - Google Sheets]

7.7 AI Marking & Teacher Review (WebView)

User Story

"As a teacher, I want to see all student submissions in a grid view with AI-generated scores per question, edit any scores directly, and release results to students."

Acceptance Criteria

- Teacher marking workstation is a grid view: rows = students, columns = questions, cells = score for that student's answer to that question.
- Column headers show question label and max marks. Max marks are drawn from the per-question marks set during assignment creation.

- Each row shows: student name, submission date/time, and AI-generated score per question.
- Student name is a link — clicking it opens the individual student submission view as a drawer for easy marking
- Assignment panel (left sidebar) lists all assignments. Each assignment shows its title and a ‘View files (N)’ link giving access to the uploaded worksheet/answer key files.
-
- On release, student receives marked scores in their webview.
- Filter on teacher review page: All / Unmarked / Marked.

7.8 Student Feedback View (WebView)

User Story

"As a student, I want to see my marked score and written feedback in a webview within my homework feed, without needing to access Learnosity."

Acceptance Criteria

- After teacher approval, student's homework item updates from 'Submitted' to 'Marked.'
- Student taps to open feedback webview: shows score from teacher.

Out of Scope

7.9 Assignment Insights (Stretch Goal)

User Story

"As a teacher, I want to see an insights summary for each assignment so that I can identify which questions students struggled with and what common misunderstandings occurred, allowing me to address gaps in the next lesson."

Acceptance Criteria

- Teacher accesses Assignment Insights via a 'View Insights' link.
- Insights open as a modal/drawer overlay anchored to the assignment context.
- Header section: assignment title and assigned date.
- Statistics section: two stat cards:
  - Submission Rate: [submitted] / [total enrolled] with completion % label (e.g. '100% completion'). Count metric, not a score.
  - Average Score: [avg score] / [max score] as raw score only. No percentage label.

- Top Performers panel: up to 3 students ranked by score (descending), showing rank, name, and raw score. Only released marks included.
- Needs Support panel: students scoring below 50% of max. If none: 'All students performing well.'
- Question Performance & Common Errors: one card per question showing:
  - Question number (e.g. 'Question 1')
  - Fraction of students who answered correctly (e.g. '2/3 students correct') — fraction only, no percentage.
  - Colour-coded status indicator (internal logic): green > 75%, red < 30%.
  - Common Misunderstanding (stretch within stretch): AI-generated one-line summary of most frequent error. Shown in muted italic style. Only shown when ≥ 2 students answered incorrectly.
- Question cards ordered by question index (Q1, Q2...), not by difficulty.
- Insights are read-only — no editing actions available.
- 'View Insights' greyed out until ≥1 submission marked and released. Tooltip: 'Insights available once marking is released.'
- Insights update in real time as additional submissions are marked and released.